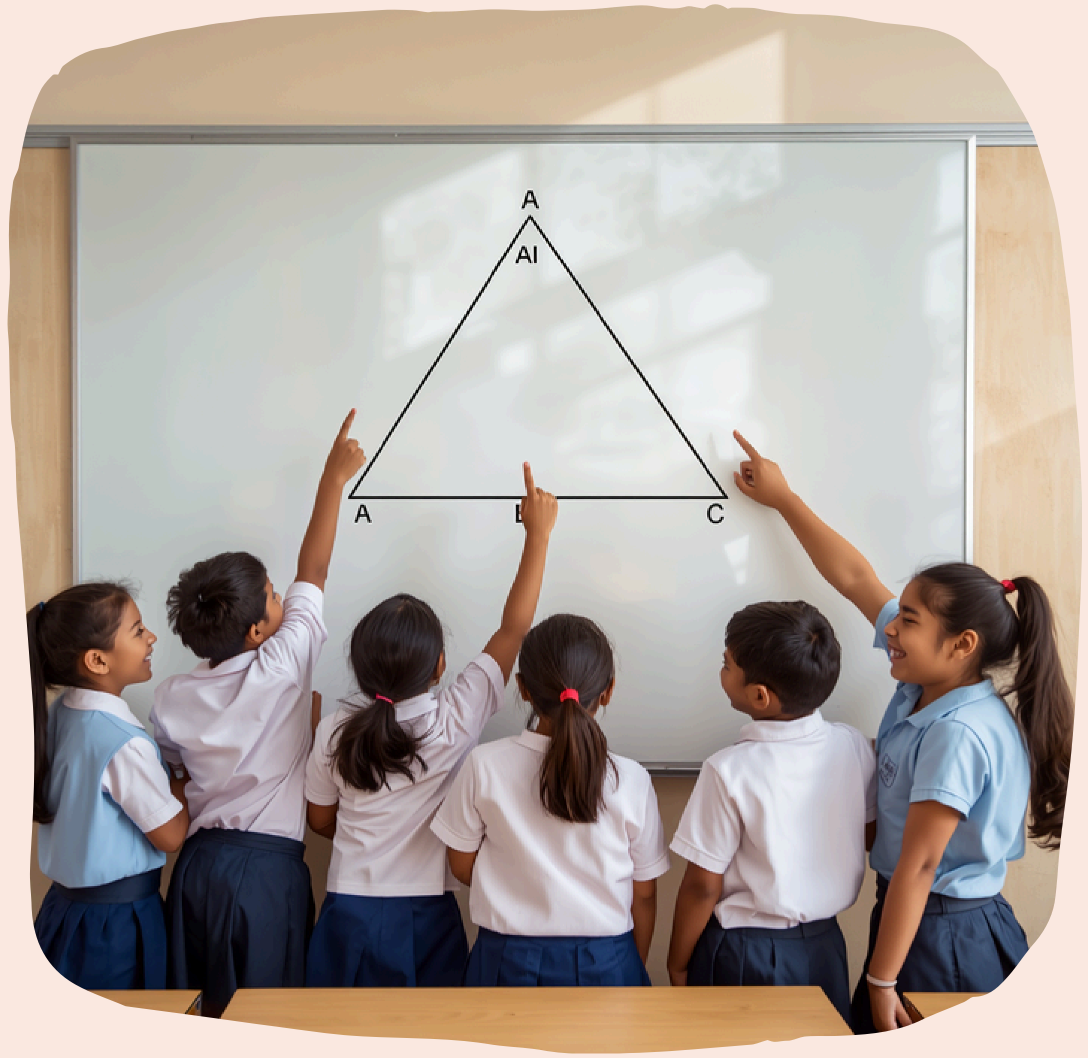
TRIGONOMETRY — SINE RULE & COSINE RULE

Learning with Fun & Curiosity!

What is Trigonometry?

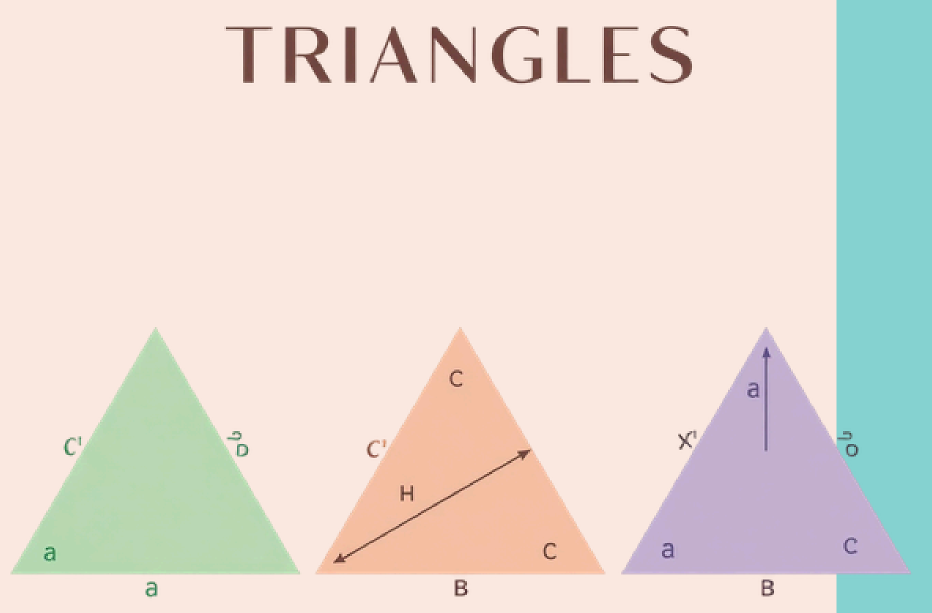


Trigonometry helps us find unknown sides or angles in triangles. It's like being a detective for shapes! When we know some parts of a triangle, trigonometry helps us discover the missing pieces.

Let's Explore!

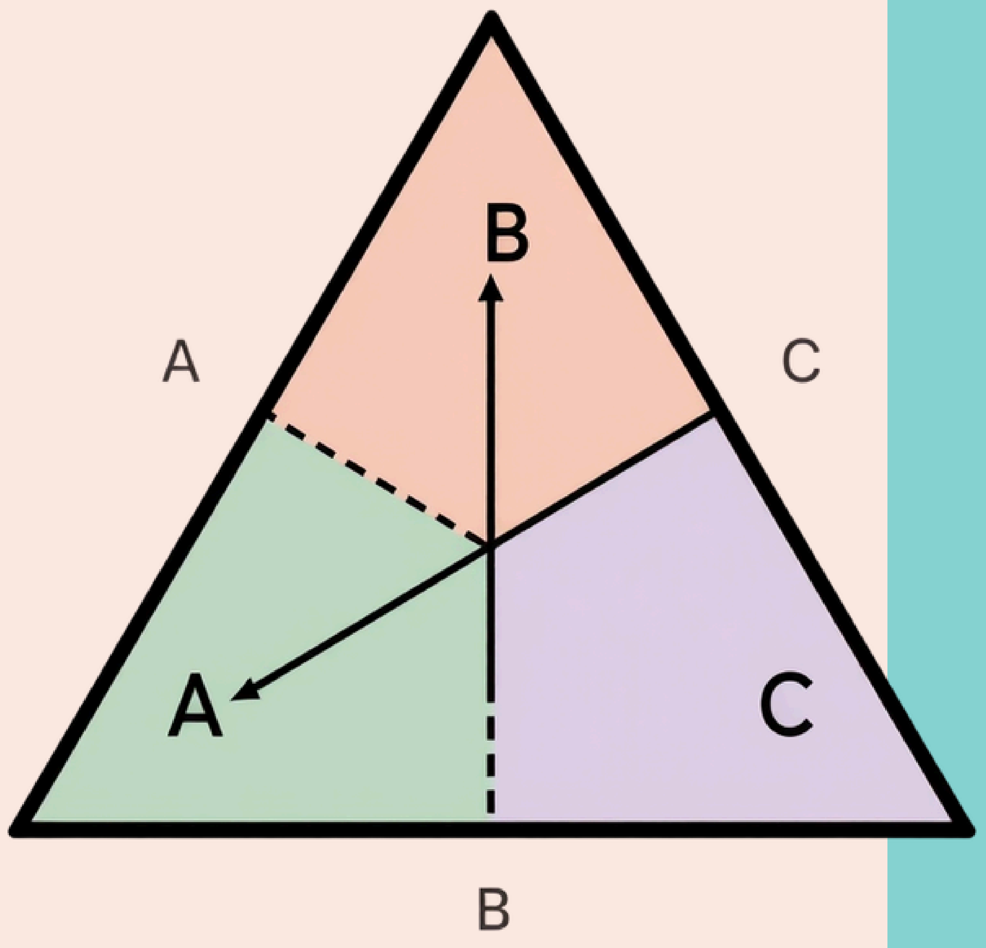
Types of Triangles & Angles

Triangles have 3 sides (a, b, c) and 3 angles (A, B, C)



Meet the Sine Rule

$$a/\sin A = b/\sin B = c/\sin C$$



Sine Rule Example

Find side b if a = 7 cm, Angle A = 50°, Angle B = 60°

When to Use the Sine Rule?

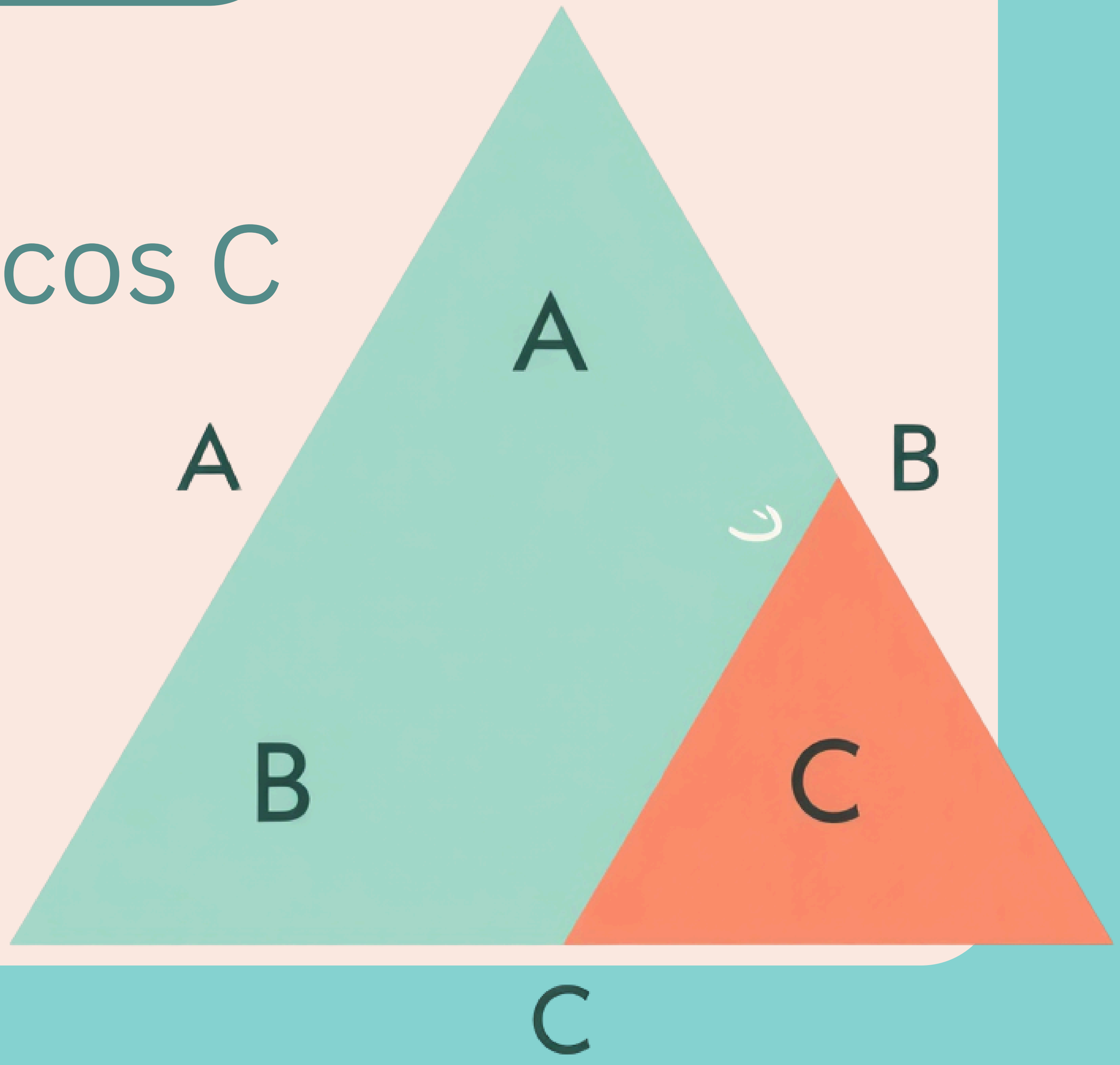
Two angles and one side known (AAS or ASA)

Two sides and an angle opposite one of them (SSA)

Remember: a/sin A = b/sin B = c/sin C

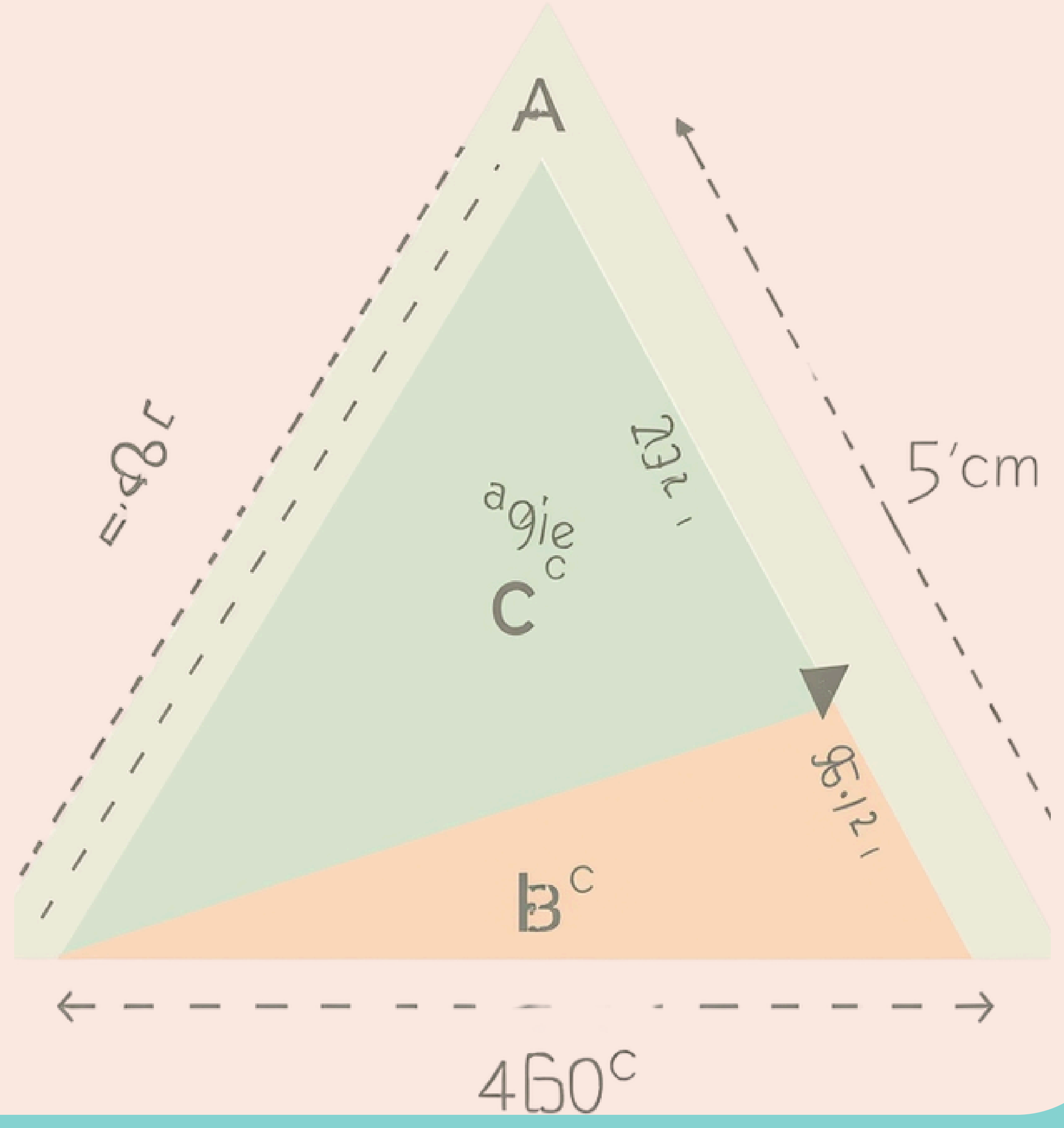
Meet the Cosine Rule

$$c^2 = a^2 + b^2 - 2ab \cos C$$



Cosine Rule Example

$$c^2 = 5^2 + 6^2 - 2(5)(6)\cos 60^\circ$$
$$c^2 = 25 + 36 - 30 = 31$$
$$c = \sqrt{31} \approx 5.57 \text{ cm}$$



When to Use the Cosine Rule?

SAS: Two sides and the included angle are known

SSS: All three sides are known to find an angle

Use cosine rule when sine rule cannot help!

Sine vs Cosine Rule Comparison

Sine Rule

- Used for AAS, ASA, SSA
- Uses ratios of sides to sines
- Formula: $a/\sin A = b/\sin B = c/\sin C$

Cosine Rule

- Used for SAS, SSS
- Uses squares of sides & cosine
- Formula: $c^2 = a^2 + b^2 - 2ab \cos C$

Choose the right rule for your triangle!

Practice Question 1

Try These Sine Rule Problems!

1. Find side b if a = 8 cm, ∠A = 45°, ∠B = 70°
2. Find ∠B if a = 10 cm, b = 12 cm, ∠A = 40°
3. Find side c if ∠A = 55°, ∠C = 65°, a = 9 cm

Practice Question 2

1. Find side c if $a = 8$ cm, $b = 5$ cm, $\angle C = 45°$
2. Find $\angle A$ if $a = 7$ cm, $b = 9$ cm, $c = 6$ cm
3. Find side b if $a = 10$ cm, $c = 12$ cm, $\angle B = 70°$

Real-life Applications of Trigonometry

Trigonometry in Action

Architecture: Engineers use sine and cosine rules to calculate roof angles and building heights.
Navigation: Sailors and pilots use these rules to find distances and directions.
Sports: Coaches measure angles for perfect throws, kicks, and jumps!

Key Takeaways

Sine Rule uses ratios: $a/\sin A = b/\sin B = c/\sin C$

Cosine Rule uses squares: $c^2 = a^2 + b^2 - 2ab \cos C$

Both rules solve any triangle, not just right-angled!

Keep Exploring Trigonometry!

Trigonometry is a fun way to explore shapes and solve puzzles! Keep practicing and stay curious!

Thank You! Any Questions?